

CCNA 200-301 Course Outline

Description

This training teaches you how to install, operate, configure, and verify a basic IPv4 and IPv6 network, configure network components, such as switches, routers, and wireless local area network (LAN) controllers (WLNs), manage network devices, and identify basic security threats. This training also covers the introduction of AI and machine learning (ML) in network operations. This Learning Path prepares you for the Cisco Certified Network Associate (**200-301 CCNA**) v1.1 exam. If passed, you earn your **CCNA certification**.

Course Objectives

The **CCNA** exam (**200-301 CCNA**) v1.1 is a **120-minute** certification exam that validates your knowledge and skills across critical networking domains. This exam covers:

- Network Fundamentals (20%)
- Network Access (20%)
- IP Connectivity (25%)
- IP Services (10%)
- Security Fundamentals (15%)
- Automation and Programmability (10%)

Skills you'll learn

1. Learn how to install, operate, configure, and verify basic IPv4 and IPv6 networks
2. Discover how to configure network components, such as switches, routers, and wireless local area network (LAN) controllers
3. Gain skills to manage network devices and identify basic security threats
4. Describe and define network programmability, automation, and SDN

Course Pre-requisites

The knowledge and skills you are expected to have before attending this training are:

- Basic computer literacy
- Basic PC operating system navigation skills
- Computer Hardware knowledge
- Basic internet usage skills
- A basic understanding of networking (*preferred but not mandatory*)

Who Should Attend

- Entry-level Network Engineers
- Entry-level Network Administrators
- Entry-level Network Support Technicians
- Entry-level Help Desk Technicians

Course Duration

- **60 Hours**, 30 Classes, 2 Hours Per Session (Online)
- **60 Hours**, 15 Classes, 4 Hours per session (Offline)
- **72 Hours**, 24 Classes, 3 Hours per session (Offline)

Module 00: Introduction to CCNA and Networking

- Introduction to Networking with Cisco
- Career Opportunities in Networking
- Industry Overview and Importance of CCNA
- Real-World Application of Networking (CCNA Scope)
- Cisco Certification Path (CCT to CCIE)
- Overview of CCNA Course Syllabus
- CCNA Exam Structure and Format
- Lab Facilities and Training Benefits

Module 01: Networking Fundamentals

- Internetworking Concepts
- Network Components (Router, L2/L3 Switch, Firewall, Access Point, Server)
- Network Types (LAN, WAN, SOHO, Cloud)
- Topologies (2-tier, 3-tier, Spine-Leaf)
- Introducing OSI and TCP/IP Model
- Data Encapsulation and De-encapsulation Process
- Network Protocols Overview
- Basic Network Design Principles
- **Quiz: Module Quiz**

Lab:

- Build a Basic LAN using Switches & PCs
- Identify Devices, Interfaces, and Cabling
- Verify Connectivity (ping, basic troubleshooting)

Module 02: Network Media & Interfaces

- Network Media Overview
- Copper & Fiber Media (Single-mode vs Multi-mode)
- Ethernet Standards & Speeds
- Wireless Standards and Technologies (IEEE 802.11)
- Cable Types (UTP, STP, Coaxial)
- Connector Types (RJ45, LC, SC)
- Power over Ethernet (PoE) Concepts and Applications
- Interface Types and Usage (Ethernet, MC, SFP)
- Network Tools and Cable Testing
- **Quiz: Module Quiz**

Lab:

- UTP Cable Crimping and Testing
- Interface Configuration and Verification
- Identify Media, Connectors, and Interfaces

Module 03: TCP/IP & Communication

- TCP vs UDP (Features and Use Cases)
- Port Numbers and Common Applications
- TCP Three-Way Handshake Process
- Data Flow, Segmentation, and Encapsulation
- IP Communication Fundamentals
- ARP Process and ARP Table
- ICMP Functions and Message Types
- Packet Flow and Traffic Analysis
- **Quiz: Module Quiz**

Lab:

- Packet Capture using Wireshark
- Analyze TCP and UDP Communication
- Inspect TCP/IP Applications and Ports
- Trace Packet Flow (End-to-End)

Module 04: IPv4 Addressing & Subnetting

- IP Address Concepts
- IPv4 Address Structure (Network & Host Portion)

- Binary and Decimal Conversion
- IPv4 Address Classes (Classful vs Classless)
- Private and Public IP Addressing
- Subnetting Techniques (FLSM)
- VLSM and CIDR
- Address Planning and Allocation
- **Quiz: Module Quiz**

Lab:

- IPv4 Subnetting Practice (FLSM & VLSM)
- Configure IPv4 Addressing on Network Devices
- Verify IP Connectivity (ping, traceroute)

Module 05: IPv6 Fundamentals

- IPv6 Introduction
- IPv6 Address Structure and Format
- Address Types (Global, Link-local, Unique Local)
- IPv6 Address Representation and Prefix
- IPv6 Configuration (Manual and Auto)
- EUI-64 Interface ID Generation
- IPv4 vs IPv6 Comparison
- Transition Techniques
- **Quiz: Module Quiz**

Lab:

- Configure IPv6 Static Routing

Module 06: Cisco IOS and Command Line Interface

- Introduction to Cisco IOS
- Cisco Device Operating Modes
- Cisco CLI Navigation and Command Structure
- Basic Device Configuration (Hostname, Banner, Password, Clock, Interface, IP Address, etc.)
- Managing Cisco Devices (Router and Switch)
- Managing Configuration Files (Running, Startup)
- Administrative Access using Console, Telnet
- **Quiz: Module Quiz**

Lab:

- Perform Basic Device Configuration
- Configure Interfaces and IP Addressing
- Configure Telnet for Remote Access
- Verify Running and Startup Configuration

Module 07: Cisco Device Access & Management

- Console Access and Initial Device Setup
- Layer 2 Discovery Protocols (CDP and LLDP)
- Secure Device Access (SSH vs Telnet)
- Local Authentication and User Management
- Remote Access Configuration (VTY Lines, SSH)
- Password Security (Enable Secret, Encryption, Policy)
- Device Hardening Best Practices
- Managing Administrative Access
- **Quiz: Module Quiz**

Lab:

- Configure Console Access and Passwords
- Configure Local User Authentication
- Configure CDP and LLDP and Verify Neighbors
- Configure SSH Remote Access

Module 08: Switching Concepts

- Ethernet Switching Concepts
- Managed vs Unmanaged Switch
- MAC Address Table (CAM Table)

- Frame Forwarding Process
- Frame Flooding Behavior
- Switching Logic and Decision Making
- Collision and Broadcast Domains
- Switching Methods (Store-Forward, Cut-Through)
- MAC Learning and Aging
- **Quiz: Module Quiz**

Lab:

- Perform Basic Switch Configuration
- Configure Default Gateway on Switch
- MAC Address Table Verification
- Troubleshoot Switch Ports and Media Issues

Module 09: VLAN & Inter-VLAN Routing

- VLAN Concepts
- Types of VLAN (Data, Voice, Management)
- Access and Trunk Ports
- DTP (Dynamic Trunking Protocol) Overview
- VLAN Tagging (IEEE 802.1Q)
- Native VLAN Concepts
- Inter-VLAN Routing Concepts
- VLAN Design and Best Practices
- **Quiz: Module Quiz**

Lab:

- Configure VLANs and Trunk Links
- Configure Inter-VLAN Routing
- Disable DTP and Configure Manual Trunk
- Verify VLAN and Trunk Operation

Module 10: STP & Layer 2 Redundancy

- Spanning Tree Protocol (STP) Concepts
- Building Redundant Switched Topologies
- Root Bridge Election Process
- Switch Port Roles and States
- Rapid PVST+ Operation
- PortFast Configuration and Use Cases
- BPDU Guard and BPDU Filter
- Loop Prevention Mechanisms
- STP Optimization Techniques
- **Quiz: Module Quiz**

Lab:

- STP Configuration and Verification
- PortFast and BPDU Guard Configuration

Module 11: EtherChannel (Layer 2/Layer 3)

- EtherChannel Overview and Purpose
- Benefits of EtherChannel
- LACP vs PAgP Protocols
- EtherChannel Load Balancing Methods
- Layer 2 EtherChannel Configuration
- Layer 3 EtherChannel Configuration
- Configuration Modes (On, Active, Passive, Desirable)
- **Quiz: Module Quiz**

Lab:

- Configure and Verify EtherChannel (LACP)
- Configure Layer 2 EtherChannel
- Verify Load Balancing and Port Status

Module 12: Layer 2 Security

- Introduction to Layer 2 Security Threats
- Layer 2 Security Features Overview
- Port Security (MAC Limiting, Sticky MAC)
- DHCP Snooping
- Dynamic ARP Inspection (DAI)
- Switch-based Attack Mitigation Techniques
- Storm Control
- Secure Switching Best Practices
- **Quiz: Module Quiz**

Lab:

- Configure and Verify Port Security
- Configure DHCP Snooping
- Configure Dynamic ARP Inspection
- Enable BPDU Guard and Verify Protection

Module 13: Routing Fundamentals

- Exploring the Functions of Routing
- Routing Table Components and Structure
- Types of Routing Protocols
- Static vs Dynamic Routing
- Longest Prefix Match (Best Route Selection)
- Administrative Distance (AD Value)
- Routing Metrics (Cost, Hop Count, etc.)
- Routing Table Codes and Interpretation
- Packet Forwarding Process
- Default Route and Gateway of Last Resort
- Routing Decision-Making Process
- **Quiz: Module Quiz**

Lab:

- Analyze Routing Table (show ip route)
- Verify Route Selection and Packet Forwarding
- Configure and Test Default Route

Module 14: Static Routing

- Types of Static Routes (Standard, Default, Floating)
- Purpose of Default Route
- Floating Static Route (Backup Route Concept)
- IPv4 and IPv6 Static Routing
- Route Summarization (Manual)
- Static Route Configuration Syntax
- Verification Commands and Techniques
- Troubleshooting Static Routes
- **Quiz: Module Quiz**

Lab:

- Configure and Verify IPv4 Static Routes
- Configure and Verify IPv6 Static Routes
- Configure Default and Floating Static Routes

Module 15: Dynamic Routing (OSPF)

- OSPF Concepts and Operation
- OSPF Neighbor Adjacency Formation
- DR/BDR Election Process
- Router ID Selection and Configuration
- OSPF States and Neighbor Relationships
- Single-Area OSPF (Area 0)
- OSPF Metric (Cost) Calculation
- OSPF Configuration Syntax
- **Quiz: Module Quiz**

Lab:

- Configure and Verify Single-Area OSPF
- Verify OSPF Neighbor Adjacency
- Analyze OSPF Routes (show ip route ospf)

Module 16: First Hop Redundancy Protocols (FHRP)

- HSRP Overview (FHRP concept)
- Active / Standby Router
- Virtual IP & Virtual MAC
- HSRP States (Init, Listen, Speak, Standby, Active)
- Priority & Preemption
- Failover Process
- **Quiz: Module Quiz**

Lab:

- Configure and Verify HSRP Operation

Module 17: Security Fundamentals (Security Core)

- Introduction to Network Security Concepts
- Threat, Vulnerabilities, Exploits, Mitigation Technique
- Security Program Elements (User Awareness, Training, Physical Security)
- Security Password Policies (Complexity, MFA, Certificates, Biometrics)
- Firewall Basics and Security Best Practices
- ACL (Access Control List) Concepts and Operation
- Standard and Extended ACLs
- Packet Filtering using ACLs
- Virtual Private Network (VPN) concepts
- IPsec Remote Access and Site-to-Site Overview
- **Quiz: Module Quiz**

Lab:

- Configure and Verify Standard ACL
- Configure and Verify Extended ACL
- Apply ACL for Traffic Filtering
- **Implementing IPsec/GRE VPN**

Module 18: IP Services (DHCP, DNS, NAT, NTP)

- Role of IP Services in Network Operations
- DNS Role and Name Resolution Process
- DHCP Operation Concepts (DORA Process)
- DHCP Client and Relay Configuration
- NAT Types (Static, Dynamic, PAT)
- NTP (Client/Server) Operation
- Functions of TFTP/FTP in the network
- **Quiz: Module Quiz**

Lab:

- Configure and verify DHCP client and relay
- Configure Static NAT
- Configure Dynamic NAT and PAT
- Configure and Verify NTP
- Backup and Restore Configuration using TFTP

Module 19: Wireless Networking

- WLAN Fundamentals and Components
- RF Fundamentals (Frequency Bands and Channels)
- Non-overlapping Wi-Fi channels
- SSID Concepts and WLAN Configuration
- Cisco Wireless Architecture (Controller-Based WLC)
- Cisco WLAN Components (AP, WLC, Link/Port, LAG)

- Access Point Modes (Lightweight, FlexConnect)
- Wireless Security (Encryption and Authentication)
- Wireless security protocols (WPA, WPA2, WPA3)
- Client Connectivity and Roaming
- **Quiz: Module Quiz**

Lab:

- Configure a Basic WLAN (SSID and Security)
- Configure a WLAN to Use WPA2-PSK
- Perform WLAN Configuration via WLC GUI (Create WLAN, Assign VLAN)
- Verify Wireless Client Connectivity
- Analyze WLAN Settings (Security, QoS, etc.)

Module 20: Network Services & Management

- Introduction to Network Management & Monitoring
- SNMP (Function and Role in Network Operations)
- Syslog (Facilities and Severity Levels)
- QoS Fundamentals (PHB, Marking, Queuing)
- Network Device Management Access Overview
- Telnet, SSH, HTTP, HTTPS, Console, Cloud-Managed
- AAA Concepts (Authentication, Authorization, Accounting – RADIUS/TACACS+ Overview)
- **Quiz: Module Quiz**

Lab:

- Configure and Verify Syslog
- Analyze Syslog Messages (Severity Levels)
- Basic QoS Classification and Marking

Module 21: Automation & Programmability

- Network Automation Overview and Benefits
- Network Programmability Concepts
- Software-Defined Networking (SDN) Architecture
- Traditional Networking vs SDN
- Controller-Based Networking (Northbound & Southbound APIs)
- REST APIs (Methods: GET, POST, PUT, DELETE)
- JSON Data Format and Structure
- Configuration Management Tool (Ansible/Terraform)
- Introducing AI and ML in Network Operations
- **Quiz: Module Quiz**

Lab:

- Basic API Testing using Postman (GET/POST)
- Analyze JSON Output from API
- Simple Automation Task (Ansible Playbook Demo)

Module 22: Final Review and Exam Preparation

- Comprehensive Course Revision (All Modules)
- Practice Questions and Exam-Oriented Quizzes
- Full-Length Mock Exams
- Exam Preparation Strategies (Theory and Lab)
- Troubleshooting Scenarios and Real-World Cases
- Time Management Techniques for the CCNA Exam
- Certification Guidance and Next Steps

Lab:

- Full CCNA Lab Scenario (End-to-End Project)
- Integrated Troubleshooting Lab (Multi-topic)

